

Carlo Russo

[LinkedIn](#) | [GitHub](#) | clo44russo@gmail.com | [Portfolio](#) | Macomb, MI | Active Secret Clearance

Open to Relocation

EDUCATION

George Washington University, Washington, DC

Doctor of Engineering, Artificial Intelligence & Machine Learning (GPA: 3.8)

Expected Dec 2026

Oakland University, Rochester, MI

M.S. Computer Science (GPA: 3.96, Summa Cum Laude)

Dec 2024

B.S. Computer Science (GPA: 3.5, Departmental Honors)

Apr 2024

SKILLS

Programming: Python, C++, SQL, TypeScript, C#

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Transformers, CNNs, RNNs, Diffusion Models

Robotics & Simulation: NVIDIA Isaac Sim, Reinforcement Learning, Robot Manipulation, Sim-to-Real Transfer, Physics Simulation, URDF

Data & ML Systems: Pandas, NumPy, Spark, Databricks, MLflow, Airflow

AI / LLM Systems: LangChain, Hugging Face, Retrieval-Augmented Generation (RAG), Vector Databases, Ollama

Cloud & Infrastructure: AWS (SageMaker, Bedrock, CloudWatch), Google Cloud (Vertex AI, BigQuery), Azure ML, Docker, Kubeflow, REST APIs

Databases: PostgreSQL, MySQL, DuckDB, SQLite

PROJECTS

Sim-to-Real Robotics Simulation Framework — Python, NVIDIA Isaac Sim, PhysX, Reinforcement Learning

- Developed a physics-based simulation environment to train reinforcement learning policies for robotic manipulation tasks and evaluate sim-to-real transfer performance.

Air Quality Forecasting System — Python, FastAPI, NewsAPI, OpenAQ API, NLP, Web Scraping

- Built a system that combines environmental APIs and news data to detect pollution events and predict upcoming air quality changes before sensor measurements reflect them.

Environmental Data Aggregation API — Python, FastAPI, REST APIs, SQL, DuckDB

- Developed a backend API that aggregates environmental data from multiple public sources, normalizes datasets, and exposes unified endpoints for querying environmental metrics.

EXPERIENCE

Selfridge Air National Guard Base: Harrison Twp, MI

Oct 2020 – Oct 2026

Aircraft Armament Systems Specialist (Drill Status Guardsman)

- Maintained and inspected aircraft armament systems supporting **8–12 fighter aircraft across scheduled drill and maintenance cycles**, ensuring compliance with strict military safety and operational readiness standards.
- Conducted **60+ equipment and weapons system inspections annually**, identifying mechanical and electrical faults to maintain mission-ready aircraft systems.
- Collaborated with **15–20 maintenance personnel across armament, avionics, and flightline teams** to execute scheduled maintenance and mission readiness operations.

Amazon: Seattle, WA

Apr 2025 – Jul 2025

Software Engineer Intern

- Developed **15+ custom CloudWatch metrics and log filters** to monitor performance of an AI-driven platform, tracking API latency, request throughput, LLM throttling, and agent utilization across **multi-region AWS environments**.
- Built **real-time monitoring dashboards tracking 10+ service KPIs**, improving observability of ML models and backend APIs for engineering teams.
- Implemented **automated CloudWatch alarms covering 20+ service conditions**, reducing manual debugging time for API performance issues by **~25–30%**.

Williams International: Pontiac, MI

Jun 2023 – Jul 2024

Software Engineer Intern

- Developed automated **test verification tools executing 100+ validation checks** to confirm engine flight code behavior against high-level system requirements.
- Implemented automated testing scripts that reduced **manual verification effort by ~30%**, improving efficiency of integration testing workflows.
- Conducted coverage testing across multiple flight software modules to support validation of **mission-critical aerospace control systems**.

CERTIFICATIONS

[DeepLearning.AI Neural Networks & Deep Learning](#)

[Career Essentials in GenAI Certificate](#)

[Google Data Analytics Certificate](#)

[IBM Machine Learning Certificate](#)

Deep Learning Specialization (In Progress)

IBM RAG and Agentic AI Professional (In Progress)